

Pouya Zarbipour-Lakposhteh

[Google Scholar](#) | [LinkedIn](#) | [Personal Website](#) | [Email](#)

Research Interests

Coastal and Ocean Engineering; Probabilistic and Uncertainty Quantification Analyses; Machine Learning and Deep Learning Applications; Remote Sensing; Extreme Events; Causal Inference; and Hydrodynamics.

Education

Tarbiat Modares University (TMU), Tehran, Iran Sep. 2021 – Feb. 2024

M.Sc. in Civil Engineering (Coastal, Port, and Marine Structures) | GPA: 15.03 (out of 20)

- Thesis Title: Probabilistic Evaluation of Effluent Discharge Performance in Desalination. | (Excellent grade 19.25/20; [Link](#))
- Supervisor: Dr. Hassan Akbari

Technical and Vocational University (TVU), Tehran, Iran Sep. 2015 – Feb. 2020

Associate's Degree and B.Sc. in Civil Engineering (Construction and Execution) | GPA: 14.94 (out of 20)

Achievements and Awards

First-ranked graduate of the Marine Structures Department, Iran 2024

Awarded for M.Sc. thesis | Department ranked 102nd in the world for civil engineering according to US News.

Governmental Fellowship, Iran Fall 2021

Awarded for M.Sc. studies | Tarbiat Modares University – Ranked 6th in Iran according to US News.

Governmental Fellowship, Iran Fall 2015

Awarded for Associate's Degree and B.Sc. studies | Technical and Vocational University

Teaching, Research and Professional Experiences

Graduate Research Assistant (under supervision of Dr. Nikoo) Jun. 2025 – Present

Sultan Qaboos University, Oman

- Developing and implementing advanced machine learning and deep learning models for ocean and water engineering applications, with a focus on predictive modeling and design decision-making.

Supervisor and Executive Engineer, Iran May. 2024 – Present

Iran Construction Engineering Organization (IRCEO)

- Supervised construction of building structures in compliance with Iran's national building regulations.

Ad Hoc Reviewer

- **Elsevier:** *Water Research* (IF:12.8, Q1); *Acta Psychologica* (IF:3.8, Q1)
- **Springer Nature:** *Scientific Reports* (IF:4.9, Q1)

Graduate Teaching and Research Assistant (under Prof. Shafieefar and Dr. Akbari) Sep. 2022 – Jun. 2025

Tarbiat Modares University and Coastal Hyd Lab

- Performing probabilistic and uncertainty quantification analyses in ocean and coastal engineering systems, supporting reliability-informed design and risk assessment.
- Preparing research manuscripts and conference contributions on ML/DL-based modeling, causal inference, and uncertainty quantification in ocean and coastal engineering.
- Teaching Assistant – *Numerical Methods in Marine Engineering*: guided students in numerical implementation of coastal and ocean engineering problems and supported assignments/projects.
- Teaching Assistant – *Design of Conventional Marine Structures*: assisted in course delivery and led hands-on training sessions in GeoStudio, PLAXIS 3D, and MIKE Zero for analysis and design of coastal and geotechnical systems.

Publications

- [J] **Zarbipour, P.**, Nikoo, M. R., Akbari, H., Nazari, R. & Karimi, M. (2026). Bridging Causality and Deep Learning for Harmful Algal Bloom Prediction. *Water Research*, 125492. <https://doi.org/10.1016/j.watres.2026.125492>
- [J] **Zarbipour, P.**, Jamali Rovesht, A., Akbari, H. & Shafieefar, M. (2026). Reliability design of reshaping berm breakwaters based on a Bayesian modification. *Ocean Engineering*. <https://doi.org/10.1016/j.oceaneng.2026.124594>
- [J] **Zarbipour, P.**, Akbari, H., & Nikoo, M. R. (2026). Quantum machine learning for wave overtopping estimation: Integrating with causal inference and uncertainty quantification. *Ocean Engineering*, 343, 123482. <https://doi.org/10.1016/j.oceaneng.2025.123482>
- [J] **Zarbipour, P.**, & Akbari, H. (2024). Reliability design of seawater desalination outfalls based on a novel probabilistic environmental assessment. *Ocean Engineering*, 313, 119465. <https://doi.org/10.1016/j.oceaneng.2024.119465>
- [J] Etri, T., Jamali Rovesht, A., Nikoo, M. R., & **Zarbipour, P.** (2026). Physics-informed neural operator with uncertainty quantification for spatial reconstruction of SWAN wave fields. *Applied Soft Computing Journal*,
- [J] **Zarbipour, P.**, Akbari, H., & SheykhBahai, A. (2025). Reliability Analysis of Berm Breakwaters based on the environmental data on the Northern coasts of Oman sea. *International Journal of Coastal, Offshore and Environmental Engineering*, e248038. <https://doi.org/10.22034/ijcoe.2026.513763.1154>
- [J] **Zarbipour, P.**, & Akbari, H. (2026). Reliability-Based Design Optimization of Berm Breakwaters with Different Reshaping and Dependency Structures. *Manuscript in preparation*.
- [J] Jamali Rovesht, A., Shafieefar, M., A., Akbari, H. & **Zarbipour, P.** (2026). Probabilistic analysis of empirical berm recession formulas considering environmental parameter dependence. *Applied Ocean Research*. <https://doi.org/10.1016/j.apor.2026.105031>
- [J] **Zarbipour, P.**, Jamali Rovesht, A., Akbari, H. & Shafieefar, M. (2026). A Reliable Machine Learning Tool for Berm Breakwater Recession Prediction. *Ocean Engineering*. <https://doi.org/10.1016/j.oceaneng.2026.128234>
- [J] Jamali Rovesht, A., **Zarbipour, P.**, Akbari, H. & Shafieefar, M. (2026). Nature Based Retrofit Solutions for Existing Berm Breakwaters: Reliability Assessment under Climate Change Impacts. *Applied Ocean Research*. Under Review.
- [J] **Zarbipour, P.**, Jamali Rovesht, A. & Nikoo, M. R. (2026). Causal-Aware Physics-Informed Neural Surrogate with Aleatoric and Epistemic Uncertainty for Wave Overtopping Prediction. *Engineering Applications of Artificial Intelligence*
- [J] **Zarbipour, P.**, Akbari, H., Nikoo, M. R. & Kazemi, A. (2026). Spatiotemporal prediction of chlorophyll-a in semi-enclosed gulfs using a hybrid graph neural network-transformer framework with satellite data and causal analysis. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-42388-0>
- [C] **Zarbipour, P.**, & Akbari, H. (2024). Assessment of the Spread of Desalination Plant Effluent Study Area: Saqi Koothar Desalination Plant, Bandar Abbas. 1st International Conference on Blue Economy. Link: <https://en.civilica.com/doc/1994723>

Project

Developed Software Tools for Coastal and Water Engineering Applications

Dec. 2021 – Present

Tarbiat Modares University and Coastal Hyd Lab

- Developed more than 10 Python-based computational tools for water and coastal engineering, including sediment transport estimators, berm breakwater design calculators, and coastal structure performance evaluators.
- Designed and developed advanced ML/DL tools—including XGBoost models for [berm breakwater recession prediction](#) and a [Causal-PINN-UQ model for wave overtopping](#)—to enhance predictive accuracy and support uncertainty-aware coastal and water engineering analyses.
- Documented and maintained the codes and user guides to support reuse by students and researchers in water and coastal engineering.

Please visit my [personal website](#) to see other completed projects.

Skills

Personal Strengths: Excellent communication and interpersonal skills; leadership and teamwork abilities; strong organizational, time, and project management skills.

Specialized Software: UQLab, MIKE Zero, SAP2000, ArcGIS Pro, Google Earth Engine, Delft3D, CORMIX.

Programming Languages: MATLAB, Python, Fortran.

Machine Learning: Supervised learning methods; tools include XGBoost and SVM; familiarity with deep learning frameworks.

Test Scores

TOEFL iBT	In progress (test scheduled)
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Reading, Listening, Writing, Speaking

References

Dr. Hassan Akbari

Associate Professor, Faculty of Civil Engineering

Tarbiat Modares University, Tehran, Iran

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Prof. Seyed Ali Akbar Salehi Neyshabouri

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